Shreyas Prasad

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Work Experience

Software Engineer Intern — Snowflake, Bellevue WA **Aug 2022 - Dec 2022**

- Worked on the storage and data lake team, maintaining the Java APIs for interfacing with the cross-cloud blob storage of all Snowflake accounts
- Rewrote 2500-line service responsible for deleting 2 billion+ expiring blob storage files per day
- Applied multi-consumer distributed queue to batch expiring file prefixes, thereby increasing file deletion throughput by 300% and eliminating the cost of old data persisting too long
- Leveraged unit, integration, and fault-tolerance testing frameworks to achieve 95+% test coverage of this new service
- Coordinated with release engineers to execute staged rollout plan of new service across 20 production deployments using parameter protection, monitoring each deployment's account using new custom metrics and alarms

Software Engineer Intern — Amazon, Toronto ON **May 2022 - Aug 2022**

- Interned on the Amazon LastMile team, responsible for building software-based transport logistics for Amazon shipments worldwide
- Created internal deployment tool to oversee the enabled features of last-mile ShipWithAmazon delivery stations, reducing team's operational toil by 75%
- Wrote internal serverless APIs for deployment tool using AWS Lambda in Java, leveraging DynamoDB tables to persist and query delivery station status
- Designed a multi-component React front-end website used by 100+ internal employees to query newly built API, and wrote CDK code to automate staging and production deployments to AWS CloudFront

Member of Technical Staff — Arctic Wolf, Waterloo ON **May 2021 - Aug 2021**

- Architected, implemented, and unit-tested concurrent Python consumer for AWS SQS-based search service receiving 1 billion+ messages per day, improving processing efficiency by 80%
- Wrote service observability alarms in Go based on Google SRE handbook, using resulting custom AWS CloudWatch metrics to detect and resolve critical application faults at high SQS message throughput
- Transferred 5 large deployments from AWS ECS to EKS, creating necessary Kubernetes configs

Projects and Leadership

Codemafia www.mafia.codes **March 2022 - Present**

- Developed real-time, full-stack Rust web game, based on the popular card game "Codenames," using Axum framework and WebSockets
- Created a light-weight, resource-efficient, channel-based game server using Rust's async programming library, Tokio
- Deployed application to Vercel using CI/CD pipeline complete with containerized test and build stages
- Built tested, and open-sourced command line tool for building type-safe Websocket messaging between Axum and Yew frameworks

Team Lead — Loo Labs . **Sep 2020 - January 2022**

- Led team of 5 student developers to build "Waterloo Wiki" for incoming students, overseeing weekly sprint activities
- Developed OIDC/SSO auth micro-server from scratch using full stack Typescript, deploying to Amazon EKS

Education

Bachelor of Computer Science, University of Waterloo, 2018-2023

- 3.8 GPA, completed 6 co-ops
- Coursework: Advanced Databases, Advanced Algorithms, Distributed Systems, Real Time Systems, Operating Systems, Object Oriented Programming, Programming for Performance

Skills

Languages: Rust, Go, Java, Python, Javascript
Tools, Technologies: AWS, Kubernetes, Docker, Shell, Databases (Mongo, Postgres), React